Posterior via Bayes: Product of Two Gaussians

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1 *Fact*

The product of two Gaussian densities (in the same variable) is proportional to a Gaussian.

So: if the prior and likelihood are Gaussian, the posterior is Gaussian, and we can read off its mean and variance by completing the square.

2 *Setup*

Let the **prior** be

$$\theta \sim N(a, R) \quad \Rightarrow \quad p(\theta) \propto \exp\Big(-\frac{(\theta - a)^2}{2R}\Big).$$

Let the **likelihood** (one observation y) be

$$y \mid \theta \sim N(\theta, V) \quad \Rightarrow \quad p(y \mid \theta) \propto \exp\Big(-\frac{(y - \theta)^2}{2V}\Big).$$

By Bayes' rule, the **posterior** is

$$p(\theta \mid y) \propto p(\theta)\, p(y \mid \theta).$$

3 *Step-by-step derivation*

3.1 *Step 1: Write the product*

$$p(\theta \mid y) \propto \exp\Big(-\frac{(\theta - a)^2}{2R} - \frac{(y - \theta)^2}{2V}\Big).$$

So the exponent (up to a sign) is

$$Q(\theta) = \frac{(\theta - a)^2}{2R} + \frac{(y - \theta)^2}{2V}.$$

3.2 *Step 2: Expand in θ*

Expand both squares:

$$(\theta - a)^2 = \theta^2 - 2a\theta + a^2, \qquad (y - \theta)^2 = \theta^2 - 2y\theta + y^2.$$

So

$$Q(\theta) = \frac{\theta^2 - 2a\theta + a^2}{2R} + \frac{\theta^2 - 2y\theta + y^2}{2V}.$$

3.3 Step 3: Collect terms in θ

Group by powers of θ:

$$Q(\theta) = \theta^2 \left(\frac{1}{2R} + \frac{1}{2V}\right) - \theta \left(\frac{a}{R} + \frac{y}{V}\right) + \left(\frac{a^2}{2R} + \frac{y^2}{2V}\right).$$

So $Q(\theta)$ is quadratic in θ: $Q(\theta) = \frac{1}{2}\sigma_{\text{post}}^{-2}\theta^2 - \theta\,\mu_{\text{post}}\,\sigma_{\text{post}}^{-2} + \text{const}$, which is the exponent of $N(\mu_{\text{post}}, \sigma^2_{\text{post}})$ up to an additive constant. We only need the coefficient of θ^2 and the coefficient of θ to identify the posterior.

3.4 Step 4: Posterior precision

The coefficient of θ^2 in $Q(\theta)$ is $\frac{1}{2R} + \frac{1}{2V} = \frac{1}{2}\left(\frac{1}{R} + \frac{1}{V}\right)$. For a Gaussian $N(\mu, C)$, the exponent is $-\frac{(\theta-\mu)^2}{2C} = -\frac{\theta^2}{2C} + \frac{\mu\theta}{C} - \frac{\mu^2}{2C}$. So the coefficient of θ^2 is $-\frac{1}{2C}$. Hence

$$\frac{1}{2C} = \frac{1}{2R} + \frac{1}{2V} \quad \Rightarrow \quad \boxed{\frac{1}{C} = \frac{1}{R} + \frac{1}{V}}.$$

So **posterior precision** = prior precision + likelihood precision.

3.5 Step 5: Posterior mean

The coefficient of θ in $Q(\theta)$ is $-\left(\frac{a}{R} + \frac{y}{V}\right)$. In the Gaussian form $-\frac{(\theta-\mu)^2}{2C}$, the coefficient of θ is $\frac{\mu}{C}$. So

$$\frac{\mu}{C} = \frac{a}{R} + \frac{y}{V} \quad \Rightarrow \quad \mu = C\left(\frac{a}{R} + \frac{y}{V}\right).$$

Using $\frac{1}{C} = \frac{1}{R} + \frac{1}{V}$, we get

$$\boxed{\mu = \frac{\frac{a}{R} + \frac{y}{V}}{\frac{1}{R} + \frac{1}{V}} = \frac{(1/R)\,a + (1/V)\,y}{1/R + 1/V}}.$$

This is the **precision-weighted average** of the prior mean a and the observation y.

3.6 Step 6: Kalman form (optional)

Let $K = \frac{R}{R+V}$. Then $1 - K = \frac{V}{R+V}$. One can show

$$\mu = a + K(y - a), \qquad C = (1 - K)R.$$

So the posterior mean is the prior mean plus K times the prediction error $(y - a)$, and the posterior variance shrinks by the factor $(1 - K)$.

4 *Summary*

- **Fact:** Prior $N(a, R) \times$ likelihood $N(\theta, V)$ in $\theta \Rightarrow$ posterior is Gaussian.
- **Posterior precision:** $\frac{1}{C} = \frac{1}{R} + \frac{1}{V}$.
- **Posterior mean:** $\mu = \frac{(1/R)\,a + (1/V)\,y}{1/R + 1/V}$ (precision-weighted average), or $\mu = a + K(y - a)$ with $K = R/(R + V)$.